

Parth Deshpande

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Education

VIT Bhopal University, Bhopal

June 2023 – June 2027

Bachelor of Technology in Computer Science & Engineering (AI&ML)

CGPA: 9.0/10

Govt. Excellence H. S. School, Betul

2023

Class 12th Board: 94%

Class 10th Board: 100%

Technical Skills

Languages: Python, C++, JavaScript (ES6+), SQL, Java

AI & Machine Learning: PyTorch, TensorFlow, Hugging Face, NLP, OpenCV, LLMs

Web Development: React.js, Node.js, Express, HTML5, CSS3, RESTful APIs

Databases: PostgreSQL, MongoDB, MySQL, FAISS (Vector DB), Firebase

Developer Tools & Cloud: Git, Docker, AWS (EC2, S3), MLflow, Postman, VS Code, Linux

Projects

Vector-Database: Custom Vector Storage Engine in C++

Jun 2026 – Jul 2026

Technologies: C++, HNSW, KD-Tree, REST API, Multithreaded I/O, JSONL Persistence

- Designed and implemented a custom vector database engine in C++, architecting three indexing strategies (HNSW, KD-Tree, Brute-Force) behind a unified query interface exposed via a lightweight REST API (cpp-httpplib).
- Optimized retrieval latency with a multilayer HNSW graph (ef_construction=200), achieving $O(\log N)$ search versus $O(N)$ brute-force scans, and persisted embeddings to disk (JSONL) for crash-safe reloads across server restarts.
- Built streaming infrastructure bridging Ollama's push-based token generation with httpplib's pull-based chunked HTTP responses, enabling real-time answer streaming and per-chunk embedding progress tracking.

VITAMIN: Microservices EdTech Backend

Aug 2025 – Mar 2026

Technologies: React-Native, FastAPI, MongoDB, Gemini API, Socket.io, Redis-style caching

- Architected the FastAPI backend and MongoDB schema for a microservices-based EdTech platform serving 20,000+ VIT students, exposing REST endpoints powering a Gemini API-driven peer-matching engine.
- Structured independent service boundaries for dashboard, planner, chat, and event-hub modules, enabling clean separation between mobile (React Native) and web clients via a shared API contract.
- Optimized data-access patterns for offline-first sync, coordinating SQLite-based client caching with server-side reconciliation via Socket.io to maintain consistency in low-connectivity environments.

Netlyze: Predictive Network Traffic Analyzer

Jan 2025 – Mar 2025

Technologies: C++, Python, scikit-learn, NumPy, Matplotlib, SQL, Wireshark

- Engineered a high-throughput backend integration layer connecting C++ packet-capture modules to a Python ML inference service, streaming live network traffic for real-time bottleneck prediction.
- Built a proactive congestion-alerting system surfacing model confidence scores and threat timelines through an interactive dashboard, reducing the interpretability gap for non-technical users.
- Processed live traffic captured via Wireshark through NumPy/scikit-learn pipelines backed by SQL storage, supporting concurrent read/write access during analysis.

Achievements

- Selected for Amazon ML Summer School 2026 — top 2.2% (~3,000) of 1.34 lakh+ registrations
- Attained Pupil Rank (Max Rating: 1409) on Codeforces
- Mastered DSA — 900+ problems on LeetCode, Codeforces & CodeChef
- Finalist, HackTU 6.0 — top team among 3,500+ students, 800+ teams

Certifications

- Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate
- Applied Data Science with Python (University of Michigan) – Coursera
- Introduction to Internet of Things (NPTEL) — IIT Kharagpur
- The AWS Technical Essentials — AWS
- Cloud Computing (NPTEL) — IIT Kharagpur